

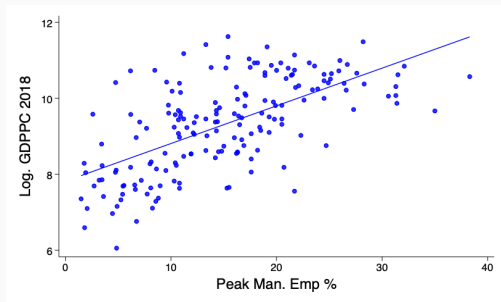
From Agriculture to Services: Development without Industrialization

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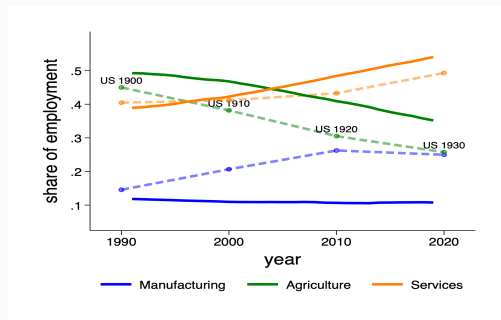
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Motivation



(a) Manufacturing and GDPPC



(b) Emp. Shares over Time

Figure 1: Scatterplot of 2019 GDPPC against peak employment share in manufacturing (left) and sector shares in 42 developing countries (right). Data Sources: Maddison (2020) and ILOEST-Gapminder (2024).

► Table

► VA

Motivation



Philippines

Income per capita: 1960: \$2,730 → 2018: \$8,730

Agricultural employment (1971): 47%

Service employment (2018):

Trade & Transport: 55% FIRE: 13%



South Korea

Income per capita: 1960: \$1,740 → 2018: \$42,700

Agricultural employment (1971): 48%

Service employment (2018):

Trade & Transport: 39% FIRE: 24%

Both countries started at similar income levels and agriculture employment shares. Only one industrialized before transitioning to services (KOR peak 28%, PHL peak 11%).

Image credits: Philippines — Kevin Moore. South Korea — Kim Hong-Ji.

Q: Can developing countries close the income gap without manufacturing? Should governments implement industrial policy to foster industrial expansion?

Mechanism:

1. Manufacturing generates demand for a skilled workforce.
2. Skilled workers generate positive spillovers by accumulating knowledge.
3. The stock of accumulated knowledge increases the human capital of future generations.
4. Economies which go through manufacturing become knowledge intensive services economies in the long-run.

Literature

- **Structural Change and Development:** Huneus and Rogerson (2023); Fujiwara and Matsuyama (2024); Spasi, Yi and Zhang (2021); Buera, Kaboski, Mestieri and O'Connell (2024), Kruse, Mensah and de Vries (2022); Rodrik (2016); Duarte and Restuccia (2010); Syrquin Chenery (1989); Kuznets (1959).
- ⇒ Add UGT to understand the consequences of premature de-industrialization on cross country income inequality.
- **Industrial Policy:** Juhász, Lane and Rodrik (2023); Chang and Andreoni (2020); Aghion, Cai, Dewatripont, Du, Harrison and Legros (2015); Rodrik (2008); Herrendorf, Rogerson and Valentinyi (2022); Peters, Zhang and Zilibotti (2023, 2026).
- ⇒ Integrate structural change in a growth model.
- **UGT:** Galor and Weil (2000); Voigtlander and Voth (2013); Doepke (2004); Hansen and Prescott (2002); Parente and Prescott (2004); Cervellati and Sunde (2015); Le Fur and Wasmer (2025).
- ⇒ Incorporate sectoral differences in models of UGT.

Data Sources

Data Sources

- **Sectoral employment shares & GDP p.c.:**
 - **ILO and Maddison via Gapminder:** for employment shares, post 1990 data is obtained from Gapminder, which harmonizes the ILOEST data; GDP p.c. is obtained from Maddison via Gapminder.
 - **GGDC:** where available, pre 1990 data is filled in from the GGDC 10-sector database.
 - **US Census of Population & Manufactures:** microdata covering the universe of population with information on employment, occupation and industry at the county level over 1850-present. Manufactures census is from Haines (2010).
- **Fertility:**
 - **IPUMS International:** microdata on fertility and employment sector for 211 country/year samples.
 - **UN via Gapminder:** aggregate fertility rates 190 countries 1800-present.
- **Education:**
 - **Barro and Lee:** 2021 update provides data on years of schooling and human capital quality for 80 high-low and middle income countries over 1960-2020.
- **Thomas Register:** Full register of manufactures producers in 1905.

Stylized Facts

Stylized Fact 1: GDP p.c. Growth Rates and Manufacturing

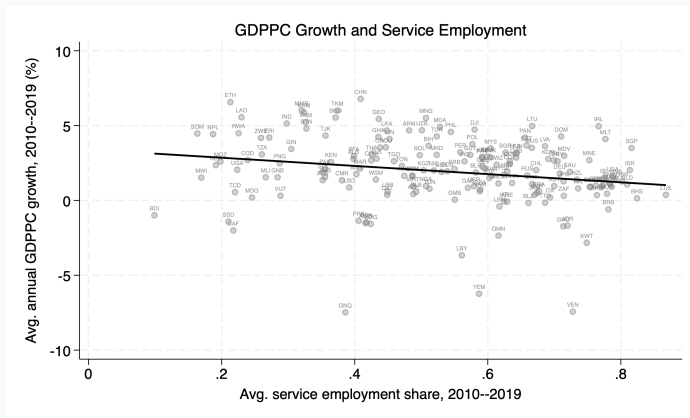


Figure 2: Baumol's cost disease. As the economy increases its service sector employment share growth slows down. Data source: ILO and World Bank via Gapminder.

Stylized Fact 1: Manufacturing GDP p.c. Growth Rates

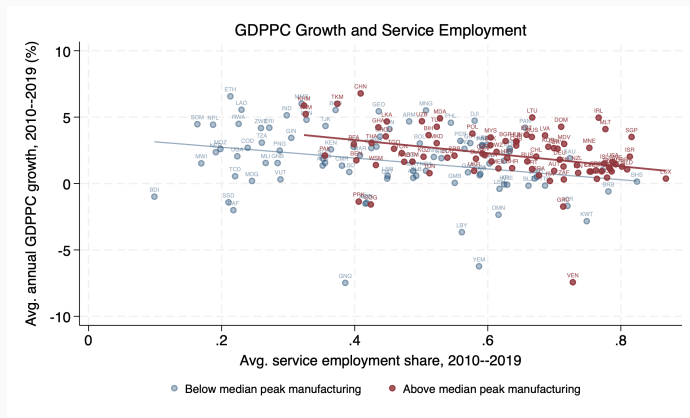


Figure 3: Baumol's cost disease. Countries with a higher peak of manufacturing employment continue to have higher growth rates. Data source: ILO and World Bank via Gapminder.

Stylized Fact 2: Manufacturing and Education

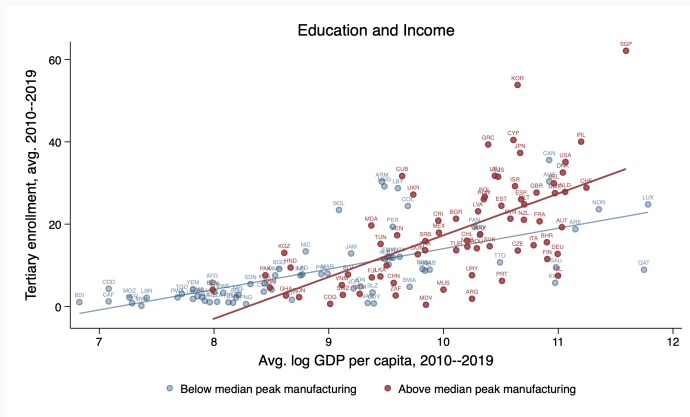


Figure 4: Tertiary enrollment rate and GDP per capita. Tertiary enrollments rise with development but more so if the economy goes through manufacturing. Data source: Barro and Lee (2015) and ILO via Gapminder.

Stylized Fact 3: Manufacturing and IO Linkages

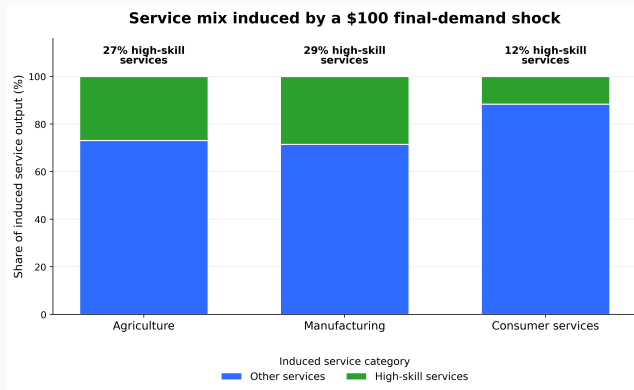


Figure 5: Bars show the service-sector mix induced by a \$100 final-demand increase in agriculture, manufacturing, or consumer services, computed from Leontief total requirements. Consumer service industries are those which supply > 50% of gross-output to final consumers. Data source: 2015 OECD Inter-Country Input-Output (ICIO)

Empirics

Using US Census data, I run two different specifications:

Long-Run Effect:

$$y_{c,s,2000} = \delta_s + \beta \text{Peak Man \% Emp}_{c,s,1850-1950} + \Gamma' X_{c,s,1850} + \epsilon_{c,s}$$

Difference-in-Difference:

$$EDSCOR50_{ct} = \delta_c + \gamma_t + \sum_{k \in \{-4,6\}} \beta_k \mathbf{1}_{\% \text{Peak Man Emp.} \in \mathcal{P} \{t - T_{\text{peak}} = k\}} + \epsilon_{ct}$$

Note: $EDSCOR50_{ct}$ is education scores within services in county c and census decade t and $\mathbf{1}_{\% \text{Peak Man Emp.} \in \mathcal{P} \{t - T_{\text{peak}} = k\}}$ is a dummy variable equal to 1 if the county is k periods away from its peak, and its peak is in quantiles $\mathcal{P}$ of the cross county distribution.

Effect of Peak % Man. Emp. on Outcomes in 2000

Table 1: Effect of Peak % Manufacturing Employment on Outcomes in 2000

	Log patents pc	Frac. college+	Log income	Innovation / total emp.	HS services / total services	Advanced mfg / total mfg
Panel A: OLS						
Peak Emp. % Man.	0.031*** (0.006)	0.127*** (0.036)	0.488*** (0.105)	0.055*** (0.021)	0.156*** (0.052)	0.080* (0.042)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1600	1600	1600	997	997	997
R2	0.192	0.317	0.401	0.226	0.133	0.300
Panel B: IV						
Peak Emp. % Man.	0.042** (0.018)	0.421*** (0.156)	0.774 (0.471)	0.190* (0.099)	0.533* (0.283)	0.038 (0.125)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap	83.91	83.91	83.91	83.91	83.91	83.91
Observations	1600	1600	1600	997	997	997
R2	0.189	0.285	0.396	0.181	0.096	0.299

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

DiD: Effect on Educational Scores Within Services

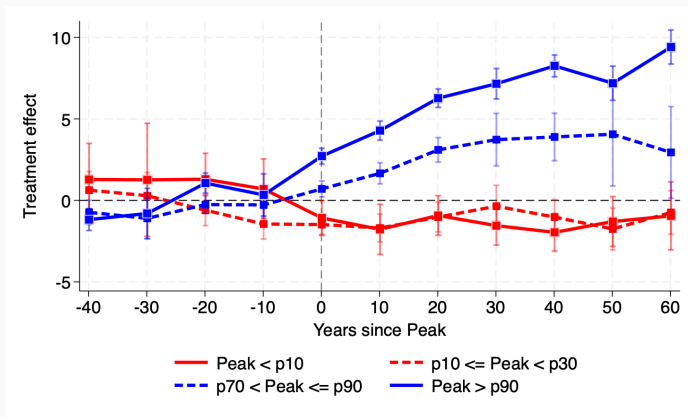


Figure 6: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k 1\{\% \text{Peak Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t ; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

Model

Households: OLG economy. Adults work, consume, choose fertility, and invest in children's education. Children are indexed by parental origin $i \in \{a, m, s\}$.

Skill formation: Children decide whether to become skilled by weighing the future skill premium against a random utility cost.

Production: There are **4 sectors**: agriculture, manufacturing, consumer services, and producer services. Producer services are a CES composite of **high-skill** and **low-skill** service tasks, with endogenous HS intensity which depends on the educational environment.

Dynamics: Fertility determines population growth, education and the skill premium determine future skill supply, and an irreversible knowledge stock H_t shifts producer-service task composition over time.

Household Problem

Parents in sector $i \in \{a, m, s\}$ choose consumption, fertility, and education:

$$\max_{\{c_{jt}\}, n_t, e_t} U_t = (1 - \gamma) \ln(C_t) + \gamma \ln(n_t h_t)$$

Consumption aggregator:

$$\sum_{j \in \{a, m, s\}} \omega_j^{\frac{1}{\sigma}} \left(\frac{C_{jt}}{C_t^{\varepsilon_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1$$

Child human capital:

$$h_{it} = \frac{e_t + \tau_h}{e_t + \tau_h + g_t}$$

Budget constraint:

$$p_{at} c_{at} + c_{mt} + p_{st} c_{st} \leq (1 - n_t(\tau_i + \tau_e(S_t)e_t)) w_t(S_t)$$

Skill choice probability:

$$p_{s,t} = \frac{1}{1 + \exp\left(-\frac{\beta_{\pi} \log \pi_t - c_0}{\sigma_{skill}}\right)}$$

where

$$\pi_t = \frac{w_{s,t}}{w_{u,t}}$$

Aggregate skill share dynamics:

$$s_{t+1} = \left(1 - \frac{1}{gen_t}\right) s_t + \frac{1}{gen_t} p_{s,t}$$

There are three final sectors: agriculture (a), manufacturing (m), and consumer services (c).

Value added in sector $j \in \{a, m, c\}$:

$$VA_{jt} = A_{jt} \left[\alpha_j (h_s L_{s,jt})^{\frac{\sigma_L - 1}{\sigma_L}} + (1 - \alpha_j) (h_u L_{u,jt})^{\frac{\sigma_L - 1}{\sigma_L}} \right]^{\frac{\sigma_L}{\sigma_L - 1}}$$

Gross output:

$$Y_{jt} = A_{go,jt} VA_{jt}^{\theta_{t,j}} M_{a,jt}^{\gamma_{a,t,j}} M_{m,jt}^{\gamma_{m,t,j}} PS_{jt}^{\gamma_{ps,t,j}}$$

$$\theta_{t,j} + \gamma_{a,t,j} + \gamma_{m,t,j} + \gamma_{ps,t,j} = 1$$

Microfoundation of PS Inputs: Task Hurdles

In sector j , producer-service tasks differ in a codification hurdle $z \geq 0$.

Primitive cross-task distribution:

$$F_j(z) = 1 - e^{-z/\kappa_j}, \quad f_j(z) = \frac{1}{\kappa_j} e^{-z/\kappa_j}$$

Interpretation of κ_j :

- κ_j is the sector-specific scale of codification hurdles.
- Higher κ_j means a thicker upper tail of hard-to-codify tasks.
- The same κ_j governs both HS task adoption and frontier codification.

Microfoundation of PS Inputs: Profit-Based Task Adoption

Producer-service tasks can be supplied in LS or HS mode.

LS mode unit cost:

$$c_t^{LS} = p_{ls,t}$$

HS mode unit cost for task z :

$$c_{j,t}^{HS}(z) = p_{hs,t} \exp\left(\frac{z}{\phi(H_t)} - 1\right)$$

A task is assigned to HS iff

$$c_{j,t}^{HS}(z) \leq c_t^{LS}$$

which implies the endogenous cutoff

$$z_{j,t}^* = \phi(H_t) \max\left\{1 + \log\left(\frac{p_{ls,t}}{p_{hs,t}}\right), 0\right\}$$

Result: Endogenous HS Task Shares

Given the sectoral hurdle distribution and the cutoff $z_{j,t}^*$,

$$x_{j,t} = \Pr(Z_j \leq z_{j,t}^*) = F_j(z_{j,t}^*)$$

So the HS producer-service task share in sector j is

$$x_{j,t} = 1 - \exp\left(-\frac{\phi(H_t) \max\left\{1 + \log\left(\frac{p_{ls,t}}{p_{hs,t}}\right), 0\right\}}{\kappa_j}\right)$$

$$p_{hs,t} = \frac{w_{s,t}}{A_{hs,t} h_s}, \quad p_{ls,t} = \frac{w_{u,t}}{A_{ls,t} h_u}$$

Result: Producer-Service Composite

Producer services are used as an intermediate input in all final sectors.

Two producer-service modes:

$$HS_t = A_{hs,t} h_s L_{s,hs,t}, \quad LS_t = A_{ls,t} h_u L_{u,ls,t}$$

Sector-specific producer-service composite:

$$PS_{jt} = \left[x_{j,t} HS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} + (1 - x_{j,t}) LS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} \right]^{\frac{\sigma_{ps}}{\sigma_{ps}-1}}, \quad j \in \{a, m, c\}$$

Thus the producer-service mix is endogenous through

$$x_{j,t} = F_j(z_{j,t}^*)$$

Update of Codification Capabilities H_{t+1}

Given the distribution of producer codification hurdles, the corresponding hazard rate is constant:

$$h_j(z) = \frac{f_j(z)}{1 - F_j(z)} = \frac{1}{\kappa_j}$$

Hence one unit of frontier-oriented educated effort in sector j yields

$$q_{j,t} = \frac{1}{\kappa_j} s_{j,t} \bar{e}_t$$

Aggregating across sectors,

$$\tilde{H}_t = \bar{e}_t \sum_{j \in \{a, m, c\}} s_{j,t} \frac{1}{\kappa_j}$$

,

$$H_{t+1} = H_t + \max\{\tilde{H}_t - H_t, 0\}$$

Population dynamics:

$$L_{t+1} = n_t L_t$$

Fertility by origin:

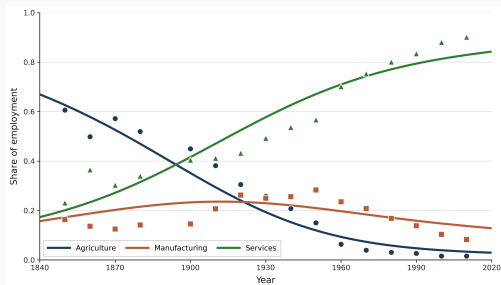
$$n_{i,t} = \frac{\gamma}{\tau_i + e_{i,t}}, \quad i \in \{a, m, s\}$$

Sectoral productivity growth:

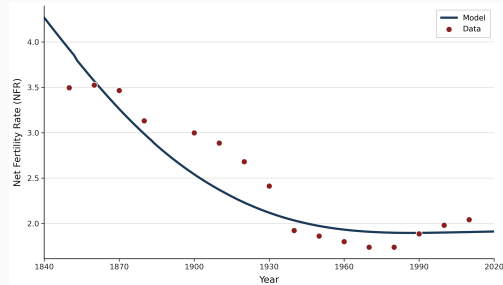
$$\frac{A_{j,t+1}}{A_{j,t}} = 1 + H_t(s_t L_t)^{\phi_j}, \quad j \in \{a, m, c\}$$

Calibration

Model Fit: USA 1850–2020



(a) Structural change



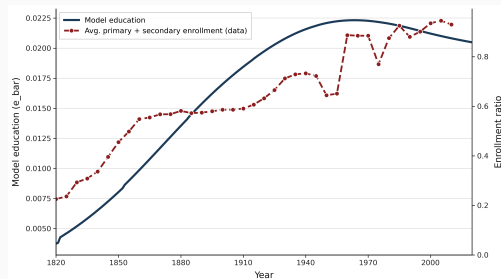
(b) Demographic transition

Figure 7: Model fit to targeted U.S. moments, 1850–2020. Dots represent data and lines show model simulations. The overall set of targeted moments includes: sector employment shares, sectoral productivity growth rates, sectoral and economy wide fertility rates, final high skill emp. share of producer services by sector, final skilled share and GDPPC levels. Data source: Gapminder.

Non-Targeted Moments: GDPPC Growth and Schooling



(a) GDPPC growth



(b) Education vs. enrollment rates

Figure 8: Non-targeted validation moments: GDP per capita growth and schooling, 1800–2020. Schooling in the model is the average education choice across sectors and in the data is the average enrollment rate (primary + secondary)/2. Data source: Gapminder and US Censuses of Population.

Non Targeted Moments: The Effect of Peak Manufacturing Emp. Share

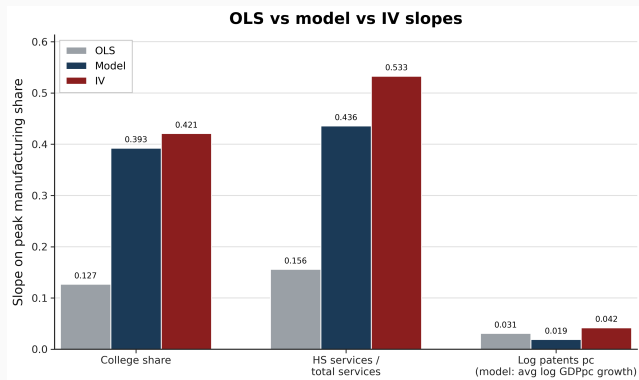


Figure 9: Comparison of the coefficients obtained from regressing key observable outcomes on the peak manufacturing employment share in the model and the data. Data slopes are consistent with Table 1.

Calibrated Parameters

Parameter	Desc.	Value	Target	Data	Model
γ	Utility weight	0.044	Avg. fertility	2.474	2.395
τ_a	Agri fert. time cost	0.419	Fert. ratio A/S	2.000	1.977
ϕ_a	Agri scale effect TFP	0.931	Agri TFP gr.	0.037	0.040
ϕ_m	Mfg scale effect TFP	0.828	Mfg TFP gr.	0.031	0.030
ϕ_s	Serv scale effect TFP	0.528	Serv. TFP gr.	0.014	0.013
g^{GO}	GO TFP growth	0.003	Avg. GDPpc dev.	0.000	0.145
κ_a	HS-task adoption friction A	1.007	HS in PS A (2019)	0.393	0.369
κ_m	HS-task adoption friction M	0.415	HS in PS M (2019)	0.653	0.683
κ_s	HS-task adoption friction S	0.731	HS in PS S (2019)	0.529	0.473
η_H	Logistic HS frontier shape	4.876	Δ Agg. HS in PS	0.182	0.168
c_{col}	College cost	0.271	Skill share (2019)	0.345	0.308
α_a	Skill share VA A	0.180	Within-VA skill A	0.160	0.161
α_m	Skill share VA M	0.254	Within-VA skill M	0.220	0.229
α_s	Skill share VA S	0.237	Within-VA skill S	0.200	0.213
ϵ_a	Agri IE	0.016	Avg. agri share	0.259	0.256
ϵ_s	Serv. IE	1.087	Avg. serv. share	0.560	0.537
w_a	Utility weight A	0.952	Agri share (1850)	0.606	0.620
w_s	Utility weight S	4.035	Serv. share (1850)	0.231	0.204

Results

Questions we can ask the model:

- Can we reduce the US' peak manufacturing employment share? What happens to its growth rate, population and GDPPC today?
 - ⇒ Change ω_m or add a wedge to the relative price of services $p_s^{Eff} = (1 + \tau_{ind})A_m/A_s$.
- How does structural change affect the demographic transition ?
- How does the model account for the cross-country facts we observe in the data? These are
 - Negative relationship between a country's peak manufacturing employment share and its growth rate.
 - Higher population growth in economies which do not go through manufacturing.
 - Higher levels of education quality (ρ) in countries with higher peaks of manufacturing activity

US Results

Results: GDPPC and Population

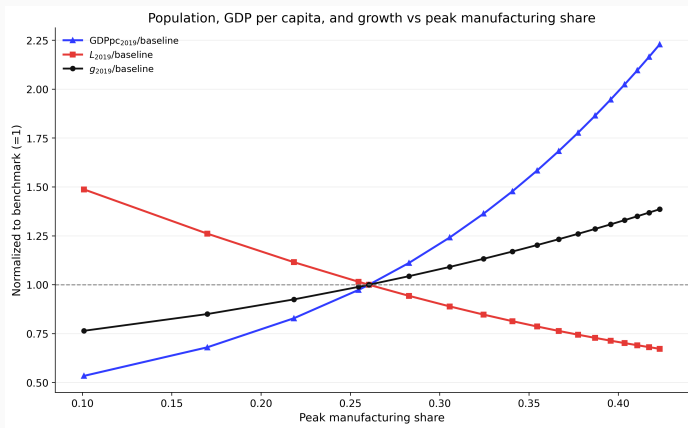
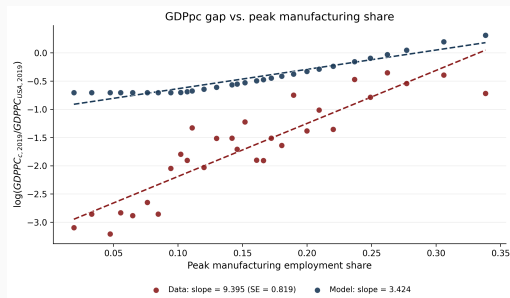


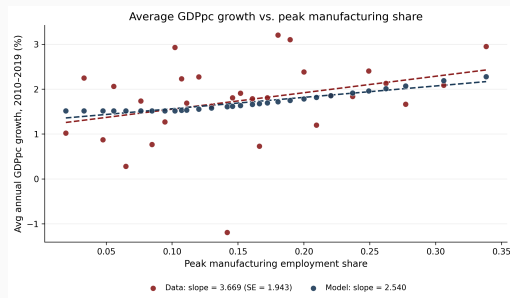
Figure 10: Comparative statics showing the effect of the manufacturing peak on outcomes in 2019 normalized at the US Benchmark values. Simulations with different peaks are obtained by shifting a preference parameter ω_m which is isomorphic to designing a value added tax schedule which reallocates labor between manufacturing and services while leaving the agricultural employment share unchanged.

Cross-Country Results

Cross Country Results: GDPPC Gaps and Growth Rates



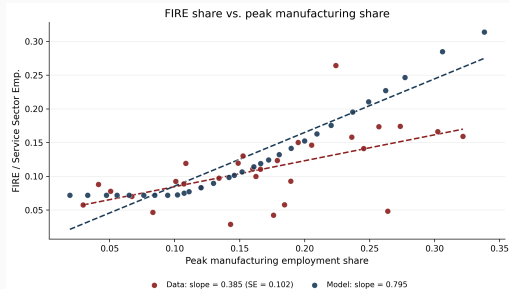
(a) $\log(GDPPC_{C,T}/GDPPC_{USA,T})$



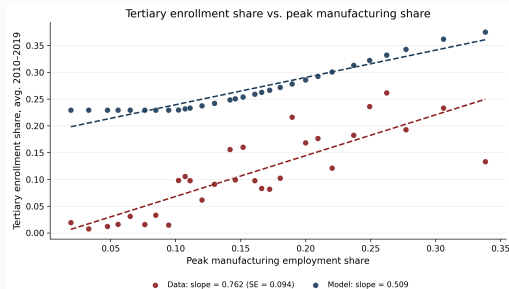
(b) $g_{C,T}^*, g_{C,T}^T$

Figure 11: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Cross Country Results: Skills and FIRE Share



(a) FIRE Share of Services



(b) Tertiary Education

Figure 12: Comparative statics showing final shares of tertiary educated workers and FIRE (Finance, Insurance, Real Estate, plus professional services) as a percentage of all service workers relative to peak manufacturing employment shares. In the model FIRE corresponds to high skill producer services.

Cross Country Results: Education and Population Growth Factors

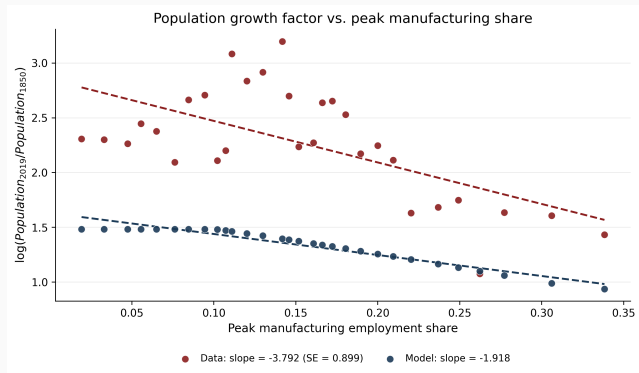


Figure 13: Binscatter plot of $\ln(\text{Pop}_{2020}/\text{Pop}_{1850})$ for economies with different peak employment shares in manufacturing.

Main idea: A growth model with structural change and endogenous fertility explains key global patterns.

- ⇒ Countries with higher manufacturing peaks $\Rightarrow$ faster growth, better education, and quicker demographic transitions.

Model exercise: Calibrate to the US (1850–2010); vary a preference shifter to simulate economies with more/less manufacturing.

Results: The model explains:

- 23.8% of GDP per capita **level gaps**
- 27.2% of GDP per capita **growth gaps**
- 53.9% of population **growth gaps**

appendix

Policy

Policy Exercises

Policy Experiments:

- Trade protection as a tool to protect industry:
 - Armington setup, small open economy (no effect on world prices). Home competes with ROW which is identical to it except having a Δ_t year head start.
 - Government implements at date T_{pol} a permanent trade openness reform, lowering its tariff from ∞ to τ^o .
- Education and fertility subsidies and taxes:
 - Government implements at date T_{pol} a proportional subsidy (or tax) to education and fertility choices which is financed through a lump sum transfer:

$$\tilde{C}_i = w_t(1 - n_i(e_i + \tau_i)) + s_e e_i + s_n n_i + B_t$$

- Industrial policy (WIP):
 - Government introduces a wedge to subsidize manufacturing employment and balances its budget by taxing services.

Trade Policy

Trade extension I: Tradable prices and demand

Tradability. Agriculture (a) and manufacturing (m) are tradable; services (s) are non-tradable.

Armington aggregation for tradables (Home). For each $i \in \{a, m\}$, Home consumes a CES composite of a Home and a foreign variety:

$$p_{i,t}^{\text{Arm}} = \left[(1 - \lambda_i)(p_{i,t}^H)^{1-\theta} + \lambda_i(\tau_o p_{i,t}^F)^{1-\theta} \right]^{\frac{1}{1-\theta}}, \quad \theta > 1.$$

$\tau_t \geq 1$ is an iceberg trade cost and $p_{i,t}^F$ is the foreign (ROW) price. Services remain non-tradable:
 $p_{s,t}^{\text{Arm}} = p_{s,t}^H$.

Trade policy. An openness instrument $\phi_o \in [0, 1]$ maps into iceberg costs τ_o :

$$\phi_t = 1 \Rightarrow \text{autarky}, \quad \phi_t = 0 \Rightarrow \text{free trade}, \quad \tau_o = \frac{1}{1 - \phi_o}$$

with liberalization happening at T_{trade} .

Trade extension II: Employment shares and dynamics

Domestic absorption under Armington. For each tradable $i \in \{a, m\}$, the domestic expenditure share is

$$\alpha_{i,t}^H = (1 - \lambda_i) \left(\frac{p_{i,t}^H}{p_{i,t}^{\text{Arm}}} \right)^{1-\theta}.$$

From consumption to employment shares. Domestic production/employment shares are:

$$\tilde{s}_{a,t}^L = s_{a,t}^C \alpha_{a,t}^H, \quad \tilde{s}_{m,t}^L = s_{m,t}^C \alpha_{m,t}^H, \quad \tilde{s}_{s,t}^L = s_{s,t}^C, \quad s_t^L = \frac{\tilde{s}_t^L}{\sum_j \tilde{s}_{j,t}^L}.$$

Key modeling choices.

- **Small open economy:** Home takes foreign prices $p_{i,t}^F$ as exogenous (from an autarkic ROW path).
- **Balanced trade:** In each period, total import value equals total export value; import absorption is proportionally scaled (real exchange rate adjustment) to satisfy $\sum_i M_{i,t} = \sum_i X_{i,t}$.

Trade Openness: Effects on Employment Shares

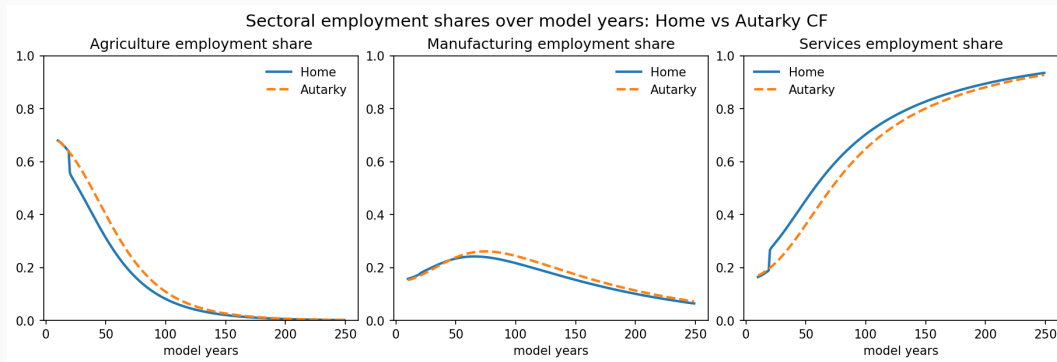
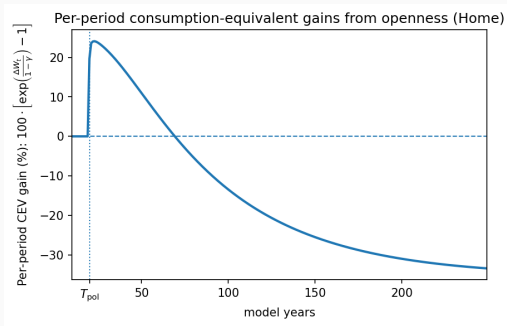
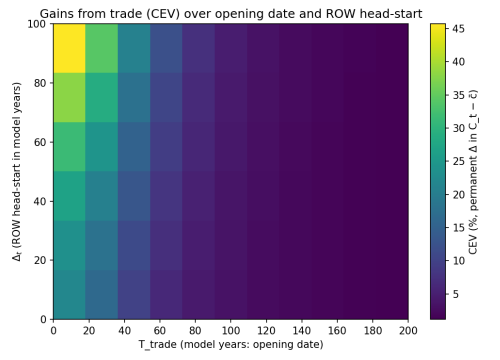


Figure 14: Trade liberalization effects. At $T_{pol} = 50$ the government permanently opens the economy to trade with a tariff of 25%. Trade openness exposes the economy to trade with *ROW* which is 50 years ahead of the home economy.

Trade Openness: Welfare Gains



(a) Gains from trade over time



(b) CEV

Figure 15: Model simulations and gains from trade. Left panel shows the percentage difference of welfare for the open economy relative to the autarky counterfactual over time. Right panel shows CEV computed using a discount rate of 2% for different openness scenarios varying the head-start of the ROW and the opening date.

Trade Openness: CEV, Growth and Manufacturing

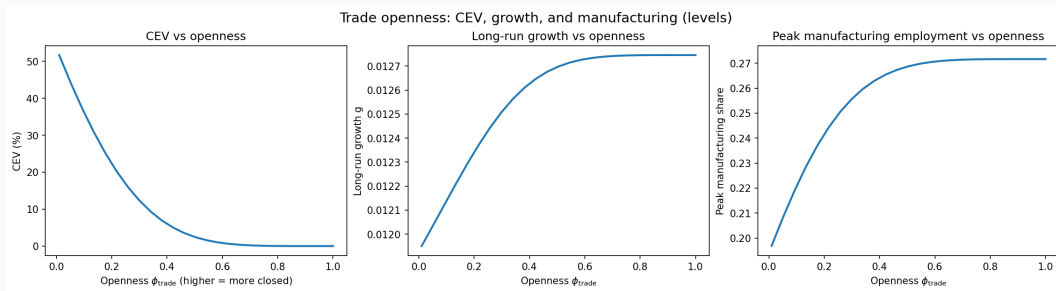


Figure 16: Trade liberalization effects. From left to right, effects on consumption equivalent variation (CEV), long run growth rate and the peak manufacturing employment share.

Trade Openness: Comparative Advantage

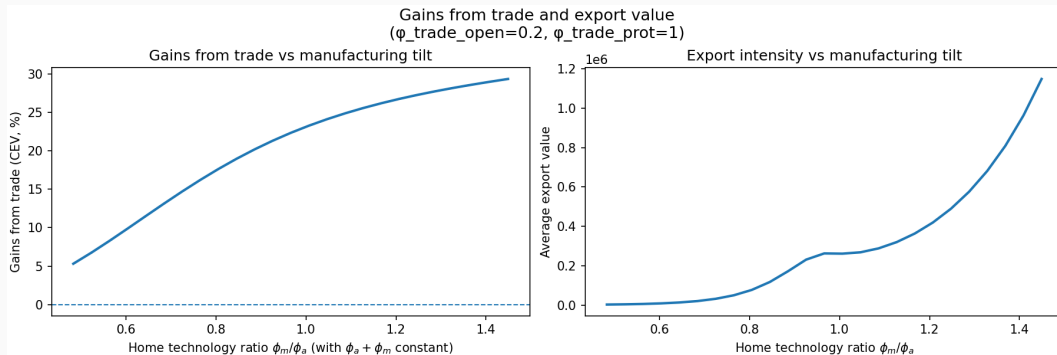


Figure 17: Trade liberalization effects for economies with different ϕ_m/ϕ_a ratios (home ϕ_m/ϕ_a is fixed at calibration values).

Education and Fertility Policy

Setup: Implementation in the Household Problem

Household budget constraint with subsidies:

$$\tilde{C}_i = w_t(1 - n_i(e_i + \tau_i)) + s_e e_i + s_n n_i,$$

where s_e is an education subsidy (or tax) and s_n a fertility subsidy (or tax).

Government budget: in every period the government balances its budget through a lump-sum tax B_t :

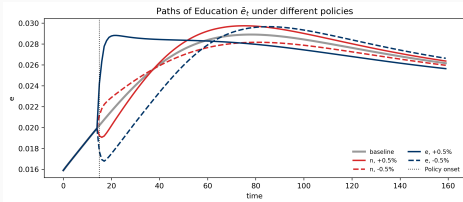
$$B_t = \sum_{i \in \{a, m, s\}} s_i(t) [s_e e_i(t) + s_n n_i(t)],$$

Actual consumption used in demand and welfare:

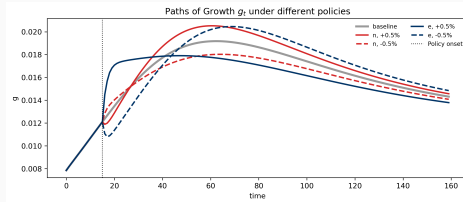
$$C_i(t) = \tilde{C}_i(t) - B_t.$$

The household does not internalize the effect that changing its decisions has on the lump-sum transfer $\partial B_t / \partial x_t = 0$ for all choice variables x_t .

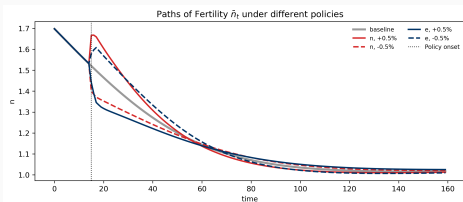
Policy Counterfactuals: Paths



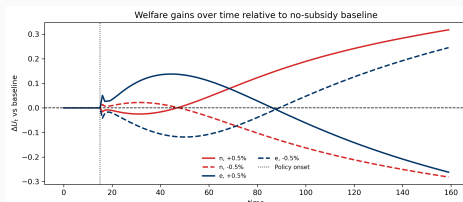
Education



Growth



Fertility



Welfare

Policy Counterfactuals: CEV

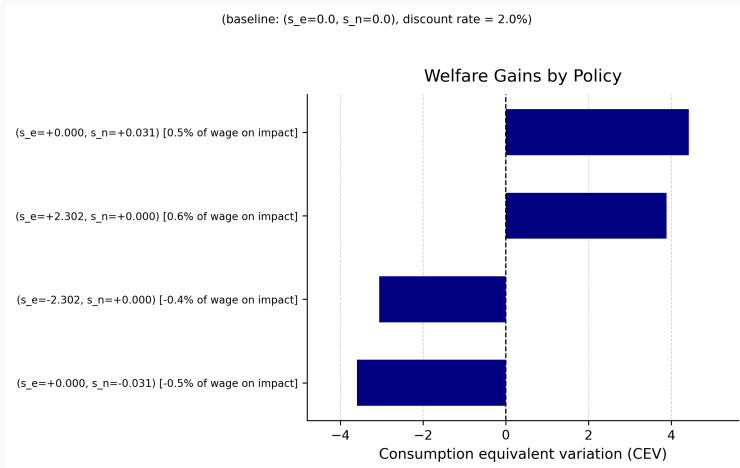


Figure 18: Ranking of policies by CEV

Conclusion

What the data show

- Higher peak manufacturing employment shares are associated with (i) faster GDPpc growth, (ii) higher education quality, and (iii) earlier fertility declines.

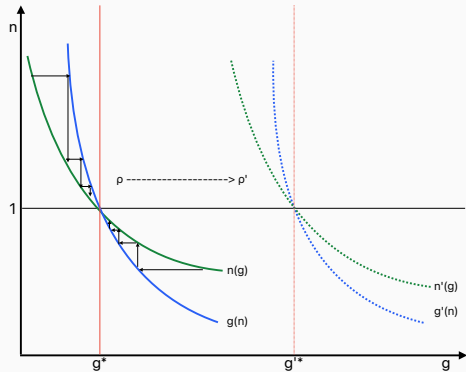
What the model does

- Structural change affects development through a **demography–human capital** channel: sectoral employment composition shifts education incentives and fertility choices, shaping long-run income levels.

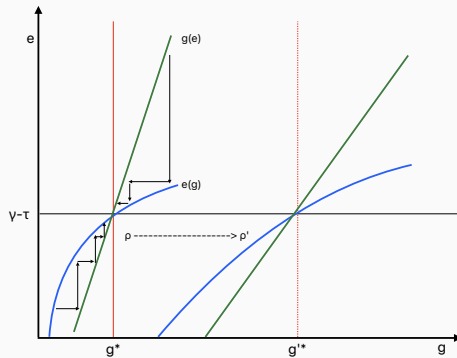
Policy implications

- **Trade policy:** openness reduces manufacturing employment share but welfare effects remain positive; gains depend on the opening date, comparative advantage and the ROW head-start.
- **Human capital policies:** education and fertility subsidies can have substantial CEV effects ($\approx 4\%$ for policies costing 0.5% of wage levels) but with different timings, fertility subsidies have gains which take a long time to materialize.

Model: Phase Diagram



(a) e and g



(b) n and g

Fact 1: Fertility and Structural Change

Table 2: OLS Regressions

	(1)	(2)	(3)	(4)	(5)
% Man. Emp.	-13.43*** (-10.74)	-4.815*** (-4.85)	-4.815*** (-4.85)	-6.113*** (-5.40)	-6.826*** (-5.54)
Log GDP p.c.		-0.808*** (-5.58)	-0.808*** (-5.58)	-0.782*** (-5.33)	0.355** (3.29)
% Agr.		0.809 (1.07)	0.809 (1.07)	0.559 (0.72)	0.0945 (0.12)
N	5053	5053	5053	5053	5053
R2	0.326	0.655	0.655	0.676	0.966
Country FE	no	no	no	no	yes
Year FE	no	no	no	yes	yes

t statistics in parentheses

Dependent variable is total fertility rate. Standard errors are clustered at the country level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Stylized Fact 1: Fertility and Structural Change

To quantify differences in fertility across sectors I run the following regression on a sample of 46 million women between the ages of 15 and 45 from 89 low and middle income countries:

$$nchild_{i,c,t} = \alpha_{a,c,t} + \delta_{e,c,t} + \gamma_{n,c,t} + \xi_{u,c,t} + \varepsilon_{i,c,t} \quad (1)$$

Where $nchild_{i,c,t}$ is the number of children for a survey respondent in country c and year t , α is a country-year-age fixed effect, δ is a country-year-education fixed effect, γ is a country-year-non-working fixed effect and ξ is a country-year-urban fixed effect.

Stylized Fact 1: Fertility and Structural Change

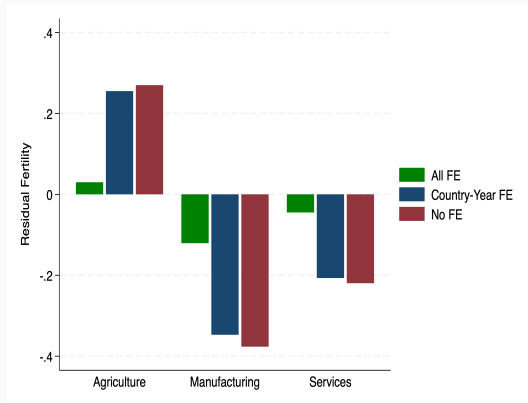


Figure 20: Average residual by sector from model 1 for different FE models. Data Source: IPUMS International.

Regression Table: GDP p.c. Growth rate in 2010-2020

Table 3: Dep. Var. is Avg GDPPC growth rate 2010-2020

	(1)	(2)	(3)	(4)
Avg. Service Employment Share, 2010–2019	-2.742*** (-2.84)	-5.171*** (-4.49)	-7.566*** (-3.59)	-9.962*** (-4.17)
Peak Emp. % Man.		9.658*** (4.59)	5.455** (2.16)	5.511** (2.09)
N	174	174	174	174
R2	0.0469	0.130	0.215	0.284
Income Decile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 3: Lucas 2009 Style Plot

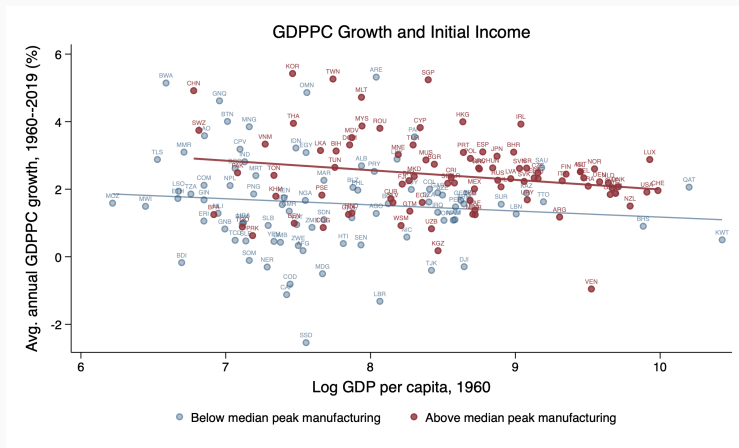


Figure 21: Scatterplot of average annual growth of GDP per capita over 1960–2020 against initial GDP per capita in 1960, grouping countries by above and below the median of peak manufacturing employment. Data source: various via Gapminder.

Stylized Fact 3: Decomposition Across Sectors

Table 4: Decomposition of Prod. Gr. Effects

	(1) Agr.	(2) Man.	(3) Serv.	(4) FIRE	(5) Non-FIRE
Peak Emp. % Man.	-0.0129 (-0.29)	0.0699** (2.25)	0.0220 (0.49)	0.133** (2.60)	0.0140 (0.27)
N	41	41	41	41	41
R2	0.00358	0.108	0.00887	0.146	0.00313

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 2: HCQ (test scores) and Peak Manufacturing

Table 5: Dep. Var. is Human Capital Quality, 2015

	(1)	(2)	(3)	(4)
Avg. log GDP per capita, 2010–2019	0.618*** (5.36)	0.523*** (2.74)	0.571* (1.82)	0.462 (1.42)
Peak Emp. % Man.	4.943*** (3.97)	4.866*** (4.05)	3.554*** (2.91)	3.237*** (2.66)
N	79	79	79	79
R2	0.552	0.559	0.697	0.702
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 2: Heterogeneity in the Service Sector

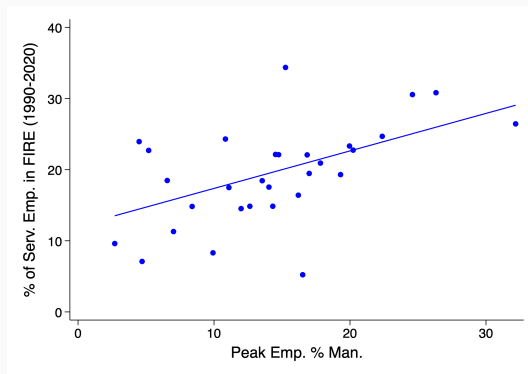
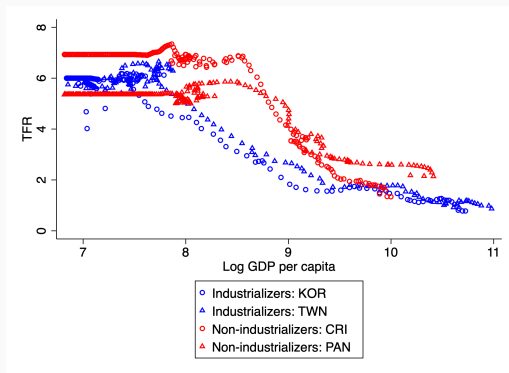
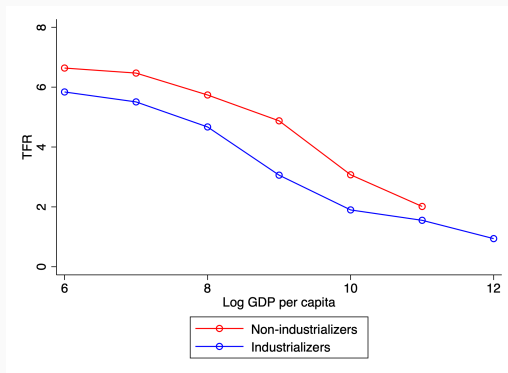


Figure 22: Binscatter plot of the fraction of service workers working in high productivity services (FIRE plus health-care and legal) against the peak employment share in manufacturing. Data source: IPUMS International for the fraction of high productivity services, GGDC for the peak employment shares of manufacturing.

Stylized Fact 1: Fertility and Structural Change



(a) Country-Comparison



(b) q1 vs q5

Figure 23: Scatterplot of the Total Fertility Rate (TFR) over log of GDP per capita in four countries. Red dots correspond to major industrializers and blue to non-industrializers. Data Source: Gapminder.

Competitive Equilibrium

A competitive equilibrium in this economy consists of household decisions c_a, c_m, c_s, n, e, i , firm decisions L_a, L_m, L_s , prices w_a, w_m, w_s, p_m, p_s , and sectoral technologies A_a, A_m, A_s . Such that, given prices:

- **Labor market:** Total labor supply equals total demand:

$$L = \sum_{i \in \{a, m, s\}} L_i.$$

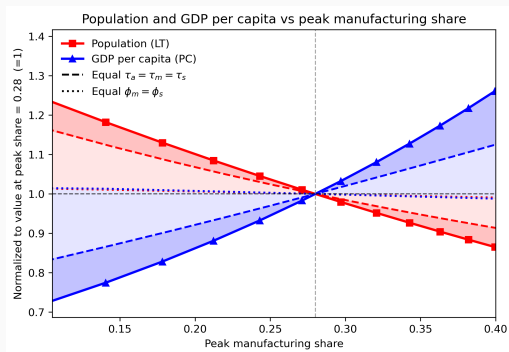
- **Goods markets:** Each sector's output equals demand:

$$y_i = c_i, \quad i \in \{a, m, s\}.$$

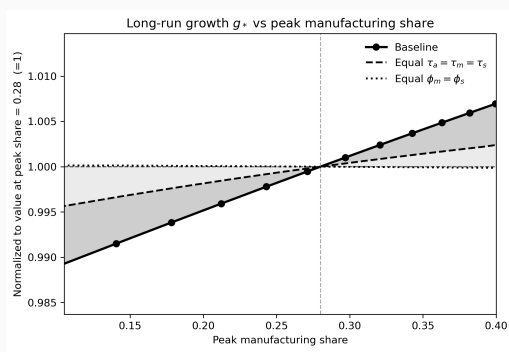
- **Optimization:** The household decisions solve the problem given in ??.

The economy is on a BGP if $g_{i,t} = g_{i,t+1} \forall i$.

Results: GDPPC and Population Decomposition



(a) $L_T, GDPPC_T$



(b) g^*

Figure 24: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Regression Table: GDP p.c. and Peak % Man. Emp.

Table 6: Dependent var is $\ln(\text{GDPPC})$

	(1)	(2)	(3)	(4)	(5)
Peak Emp. % Man.	9.308*** (11.80)	8.898*** (9.69)	3.182*** (3.91)	4.620*** (4.97)	1.950*** (2.63)
Years Since Peak		0.00595 (1.26)	-0.00136 (-0.36)	0.0103** (2.23)	0.00104 (0.28)
N	176	176	176	176	176
R2	0.426	0.430	0.766	0.605	0.812
Region FE	no	no	no	yes	yes
WB Institutions Controls	no	no	yes	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fertility and Education Choices

Define $\tilde{w} \equiv \frac{\bar{c}}{1-\gamma}$, the level of w such that if the household allocates $(1 - \gamma)$ of its resources to consumption, it can just satisfy the subsistence constraint. Then for all households i :

$$n_{it}(\tau_i + e_{it}) = \begin{cases} \gamma & \text{if } w_t \geq \tilde{w}, \\ 1 - \frac{\bar{c}}{w_t} & \text{if } w_t \in (\bar{c}, \tilde{w}), \\ 0 & \text{if } w_t \leq \bar{c} \end{cases} \quad (2)$$

Implications for effective resources:

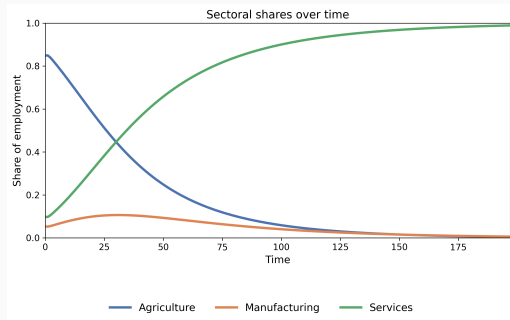
$$\underbrace{(1 - n_{it}(\tau_i + e_{it}))}_{R_t} w_t = \begin{cases} w_t(1 - \gamma) & \text{if } w_t \geq \tilde{w}, \\ \bar{c} & \text{if } w_t \in (\bar{c}, \tilde{w}) \end{cases}$$

Optimal education choice:

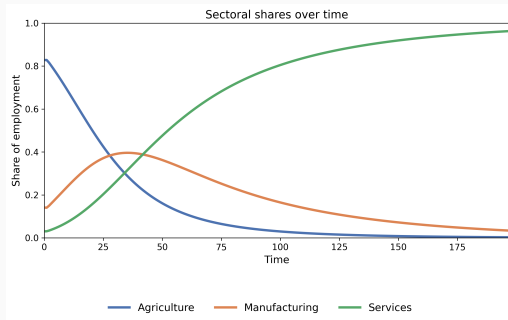
$$e_{it}^* = \max \left\{ 0, \sqrt{\bar{g}_t(1 - \rho_t)\tau_i} - \rho_t\tau_i \right\}$$

Model: Qualitative Patterns 1/2

Consider two economies with different ω_m :



(a) $\omega_m = 0.5$



(b) $\omega_m = 3.5$

Model: Qualitative Patterns 2/2

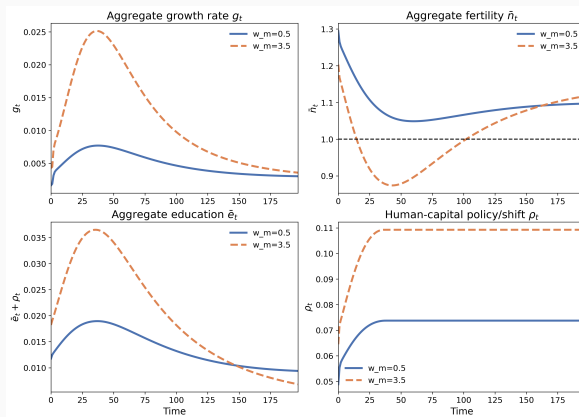
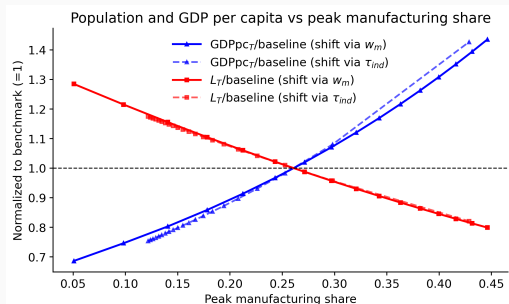
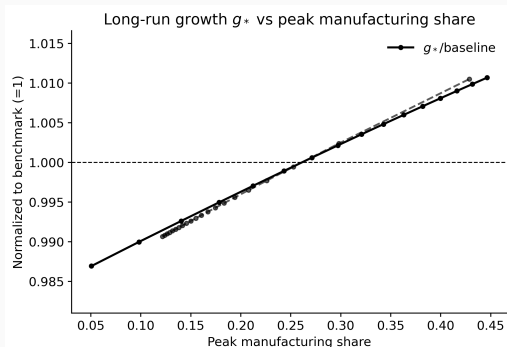


Figure 26: Comparison of Dynamics for two economies. The blue line corresponds to the economy which does NOT go through manufacturing (low peak), and the orange line the economy which does go through manufacturing (high peak).

Results: GDPPC and Population Policy Shifter vs ω_m



(a) $L_T, GDPPC_T$



(b) g^*

Figure 27: Final GDPPC and Population for economies with different peak employment shares in manufacturing. Comparison between counterfactuals obtained by changing the utility weight on manufactured goods ω_m , vs a policy wedge τ_{ind} which affects the relative price of services.

Table 7: Non-calibrated parameters (fixed and normalizations)

Parameter	Value	Source / normalization
ν	0.5	Cobb-Douglas (normalization)
σ	0.5	CES elasticity of substitution (standard)
ϵ_m	1	Manufacturing income-elasticity (normalization)
$\bar{r}$	0.05	Exogenous interest rate (normalization)
ω_m	3.5	Manufacturing utility weight (normalization)
T^{gen}	20	Generation length in years (Lagerlof, 2006)
h_0	1	Initial human capital (normalization)
A_0	[1.0, 1.0, 1.0]	Initial TFP levels (normalization)
$\bar{c}$	0.1	Subsistence / minimum consumption (fixed)

Residual Fertility by Sector: USA 1850-2020

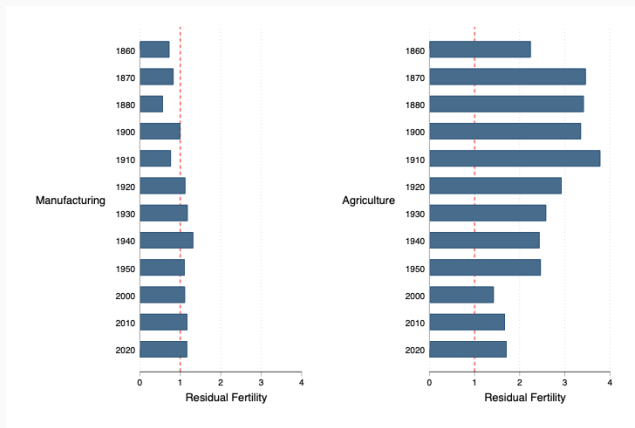


Figure 28: Sectoral fertility relative to services. Residuals from model 1 using samples of 25-35 year old women from the US census of population are computed and then normalized by the service sector fertility. For this exercise the model only includes a year fixed effect. Bars show the average residual in a year across sectors. Data source: IPUMS USA.

Residual Fertility by Sector: USA 1850-2020

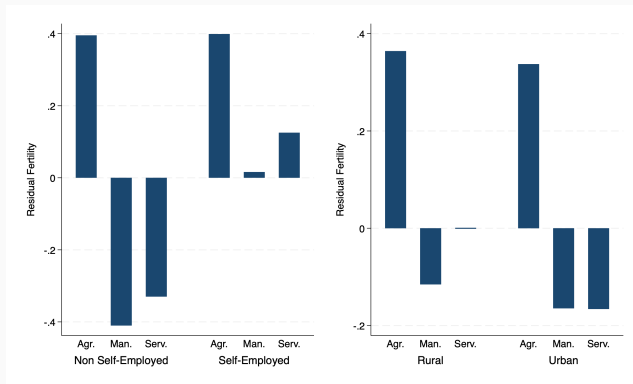


Figure 29: Residuals from model 1 using samples of 25-45 year old women from IPUMS International. For this exercise the model only includes a country-year fixed effects. Bars show the average residual in a year across sectors. Data source: IPUMS International.

Table: Decomposing the Demographic Transition

Table 8: Total fertility decline (1850–2010): data and model counterfactuals

Scenario	Decline ($NFR_{1850} - NFR_{2010}$)	% of data decline
Data	1.453	100.00%
Model (no SC)	0.483	33.25%
Model (no education increase)	1.295	89.16%
Model (total)	1.548	106.57%

Calibration: GDP p.c.

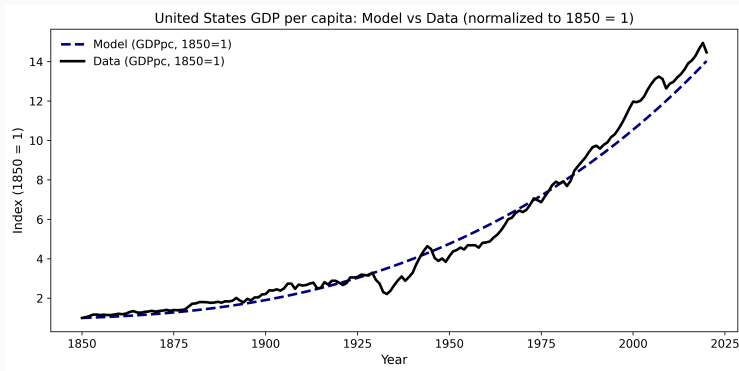


Figure 30: Calibration of GDP per capita over the development process. The solid black line shows data GDP per capita in the US over 1850-2020, the dashed blue line shows the model path. Data source: Maddison (2020).

Calibration: Relative Productivity

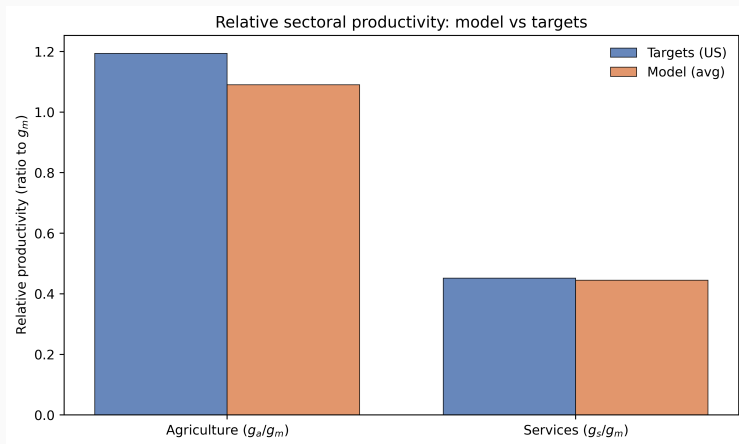


Figure 31: Relative productivity targets. Blue bars correspond to the relative productivities of agriculture and services computed using the GGDC 10-sector database, red bars show the equivalent in the model. Data source: GGDC 10-sector.

Calibration: Sectoral Fertility

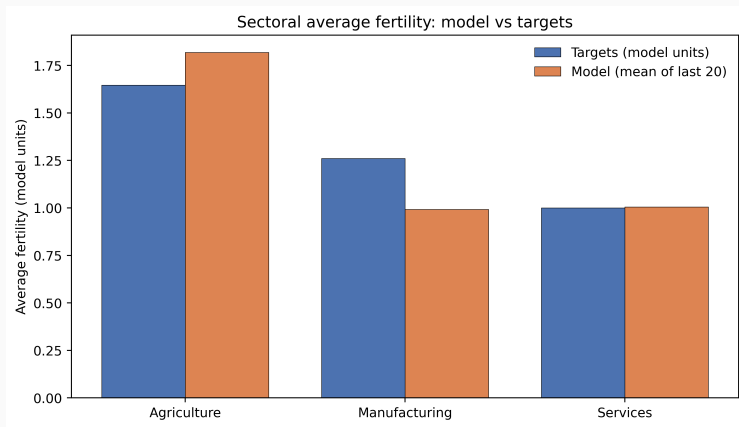


Figure 32: Sectoral Fertility: model vs targets. Blue bars correspond to the average fertility of agricultural, manufacturing and services households (assuming replacement fertility in services) in model units using the last two samples of IPUMS data. Red bars show the corresponding values in the model. Data source: IPUMS International.

CEV Formula

Letting W_t^{open} and W_t^{aut} denote per-period welfare under openness and protection, respectively, the CEV λ solves

$$\sum_{k=0}^{T-1} \beta^k (W_{t_0+k}^{\text{open}} - W_{t_0+k}^{\text{aut}}) = (1 - \gamma) \ln(1 + \lambda) \sum_{k=0}^{T-1} \beta^k,$$

where β is the discount factor per model period and $1 - \gamma$ is the weight on the consumption term in utility.

Solving for λ yields

$$\lambda = \exp\left(\frac{\sum_{k=0}^{T-1} \beta^k (W_{t_0+k}^{\text{open}} - W_{t_0+k}^{\text{aut}})}{(1 - \gamma) \sum_{k=0}^{T-1} \beta^k}\right) - 1.$$

CEV: Agriculture vs Manufacturing Comparative Advantage

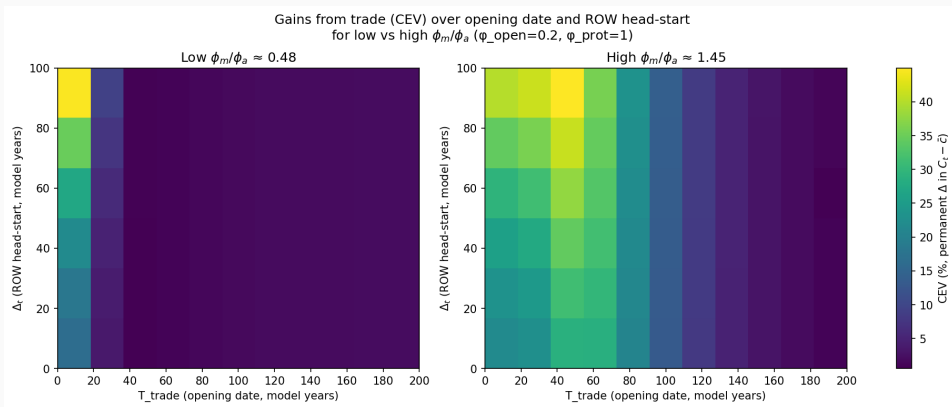


Figure 33: Welfare gains from trade openness over different opening dates and ROW head-start in two economies with different ϕ_m/ϕ_a ratios. The economy with a high ϕ_m/ϕ_a has a comparative advantage in manufacturing over ROW, where the ϕ_m/ϕ_a is ≈ 1 . The economy with a low ϕ_m/ϕ_a has a comparative advantage in agriculture.

Motivation: Value Added

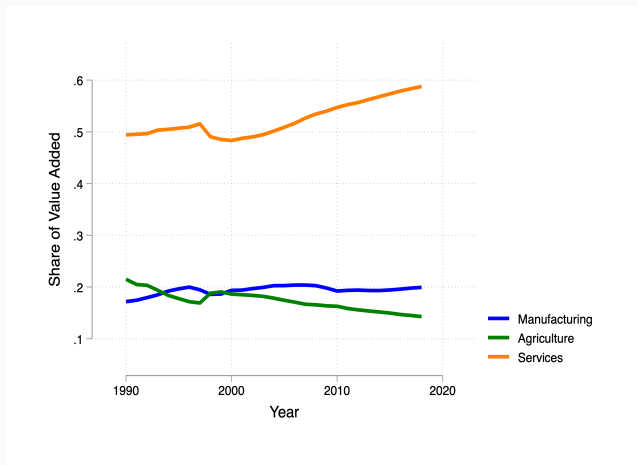


Figure 34: Value added shares over 1990-2018 compared to the US over 1900-1930. Data source: US Manufacturing Censuses and and GGDC (2023).

DiD: Constructing a Newspaper Labor-Demand Panel (1850–1940)

Step 1: Scrape universe of relevant pages (Chronicling America). Baseline data is from O'Rourke, Rapone and Santaaulalia-Llopis (2026), which uses a proximity search for “*help wanted*” (within 5 words) and downloads all $N \approx 550k$ pages from the LOC public repository.

Step 2: Build a stratified sample: I draw a *stratified random sample* at the newspaper level. $N \approx 110k$.

Step 3: Map newspapers to regions. I geo-locate each newspaper and cluster locations into $K = 200$ geographic regions using K-means on coordinates.

Step 4: Construct region-year outcomes from text counts. I count keyword hits for: (i) skilled manufacturing occupations (e.g., machinist, millwright, boilermaker, ...); (ii) skilled services/white-collar occupations (e.g., clerk, bookkeeper, stenographer, ...); (iii) a baseline basket: *cook + waiter + janitor + laborer*.

Step 5: Define treatment timing from “new/expanding factory” triggers. A region's treatment year is the first year with persistent mentions of new/expanding manufacturing establishments (threshold: ≥ 5 trigger pages in that region-year; restricted to 1880–1930).

Help Wanted Ads: Descriptive Results

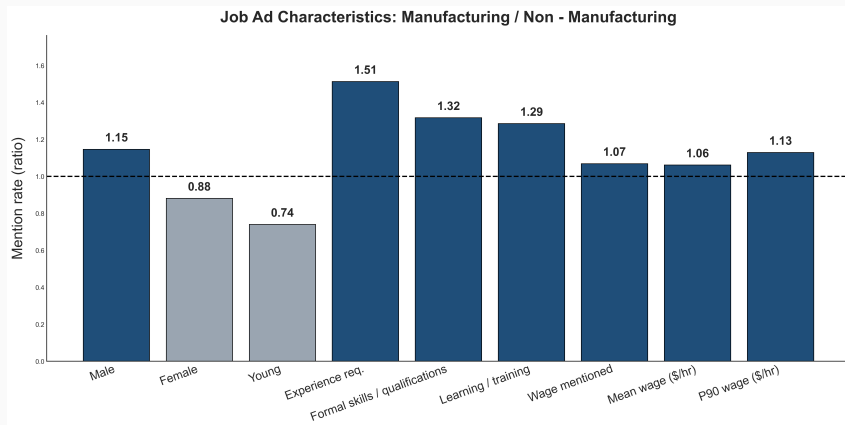


Figure 35: Job-ad characteristics in manufacturing vs. non-manufacturing contexts from historical U.S. newspapers (Chronicling America). An ad is classified as “manufacturing” if manufacturing context terms appear within ± 15 words; otherwise it is “general.” Each bar shows the relative mention rate of gender, age, experience/qualification and training related terms. When present, wages converted to hourly equivalents. Data source: Author’s calculations using LOC “Chronicling America”.

DiD: Event Study Effect of "New Factory" on Demand for Skilled Labor.

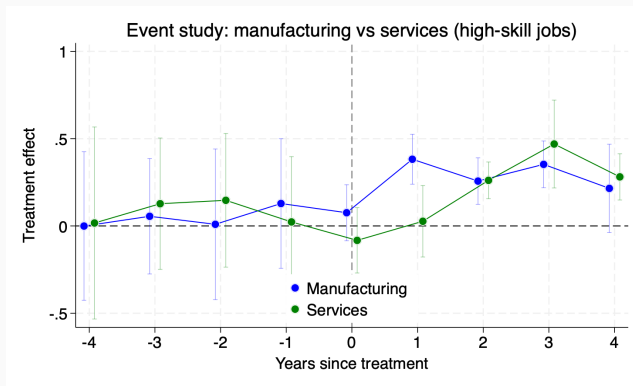


Figure 36: Event-study coefficients from Borusyak et al.'s imputation estimator using region-year newspaper data. The dependent variable is the log of occupation keyword hits normalized by a baseline basket of unskilled professions (cook, waiter, janitor, laborer). Vertical bars show 95% confidence intervals clustered by region. Data source: Author's calculations using LOC "Chronicling America".

DiD: Event Study Effect of "New Factory" on Demand for Skilled Labor Long Run

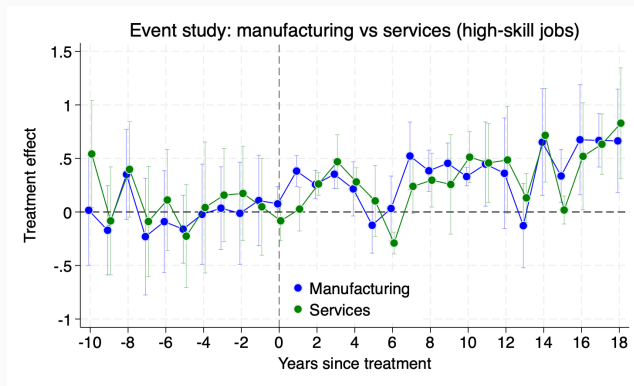


Figure 37: Event-study coefficients from Borusyak et al.'s imputation estimator using region-year newspaper data. The dependent variable is the log of occupation keyword hits normalized by a baseline basket of unskilled professions (cook, waiter, janitor, laborer). Vertical bars show 95% confidence intervals clustered by region. Data source: Author's calculations using LOC "Chronicling America".

Universities Across US Counties

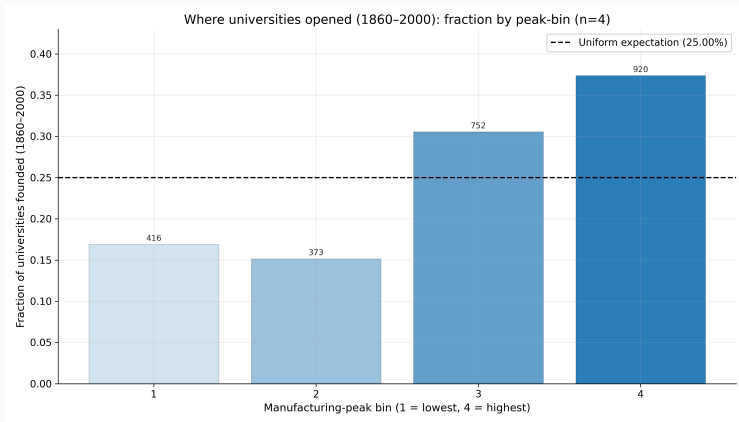


Figure 38: Fraction of universities founded between 1860 and 2000 across counties grouped into 4-quantiles of historical peak manufacturing intensity. Data on university openings is scraped from Wikidata.

DiD: Two Different Service Sectors (Industry Breakdown)

Bottom 10		Top 10	
Sub-industry	%	Sub-industry	%
Private households	3.5	Educational services	75.8
Shoe repair	4.3	Legal services	73.5
Trucking	5.3	Medical (excl. hospitals)	71.2
Taxicabs	5.7	Welfare & religious	59.1
Dressmaking	6.1	Engineering & architecture	58.4
Rail & bus transit	6.4	Misc. professional	53.9
Misc. repair	7.1	Drug stores	45.7
Sanitary services	7.2	Radio & TV	43.0
Auto repair	8.0	Accounting & auditing	42.7
Water transport	8.1	Hospitals	38.5

Note: Shares indicate the fraction of individuals in each sub-industry who completed one or more years of college, 1950 Census.

DiD: Two Different Service Sectors (Occupation Breakdown)

Bottom 10 (excl. manufacturing)		Top 10	
Occupation	%	Occupation	%
Teamsters	0.0	Mathematicians	100.0
Boatmen/canalmen/lock keepers	0.0	Biological scientists	96.0
Bootblacks	0.0	Chiropractors	93.9
Telegraph messengers	0.0	Professors/instructors (unspecified)	93.8
Midwives	0.0	Professors: Engineering	93.8
Hucksters & peddlers	1.7	Professors: Chemistry	93.8
Longshoremen & stevedores	2.1	Professors: Biological sciences	93.8
Laundresses (private household)	2.2	Professors: Mathematics	93.8
Charwomen & cleaners	2.7	Professors: Agricultural sciences	93.8
Private household workers (n.e.c.)	2.9	Professors: Physics	93.8

Note: Shares indicate the fraction of individuals in each occupation with ≥ 1 year of college (1950 Census). Manufacturing-heavy occupations (e.g., textile/metal/printing shop trades, mill/plant operatives, mining/industrial production jobs) are excluded from the bottom list.

DiD: Effect on Educational Scores (more groupings)

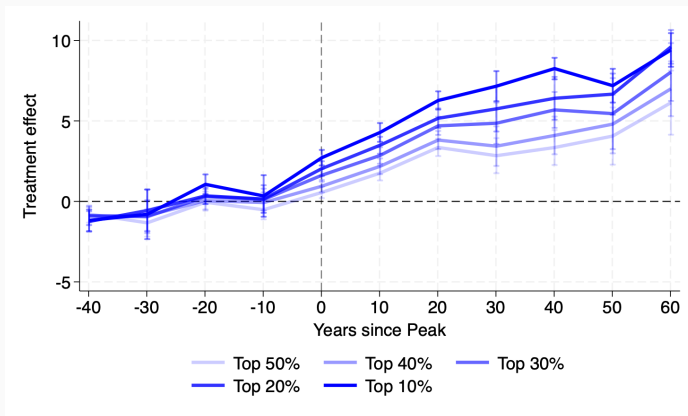


Figure 39: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k 1\{\% \text{Peak Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t ; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

Map: Geography and the Peak

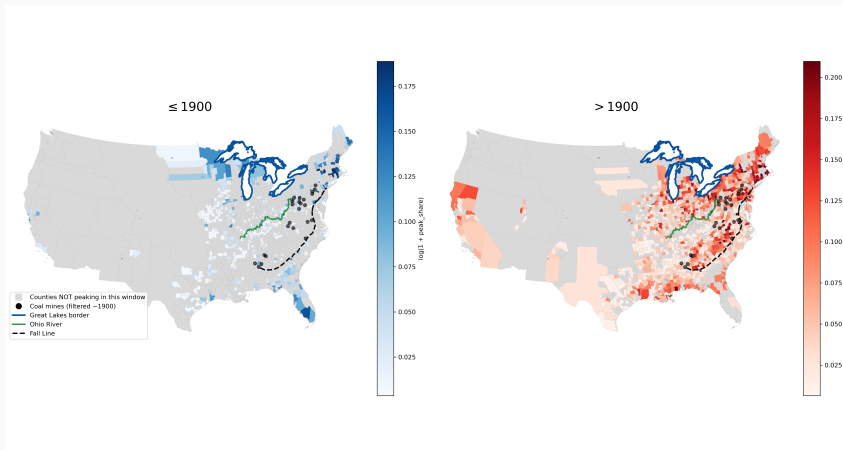
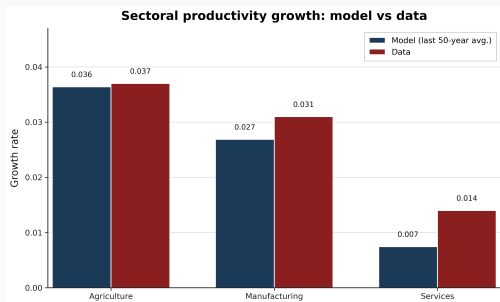
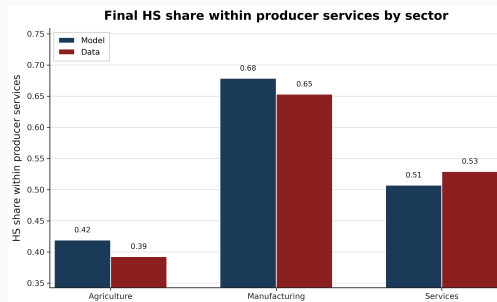


Figure 40: Map of peak manufacturing employment shares in the pre and post 1900 time periods. Data source: Author's calculations using Haines (2010), NHGIS and USGS.

Model Fit: Productivity and HS Shares



(a) VA productivity growth rates



(b) HS shares in producer services

Figure 41: Model vs Data. Panel (a) shows the productivity growth rates of value added in the model and in the data, panel (b) shows the final share of high skill employment within producer services in the model and the data. Data sources: value added productivity is from the GGDC 10-sector database, HS employment shares within producer service occupations is from the US 2019 Census of Population.

Model Fit: GDPPC Levels

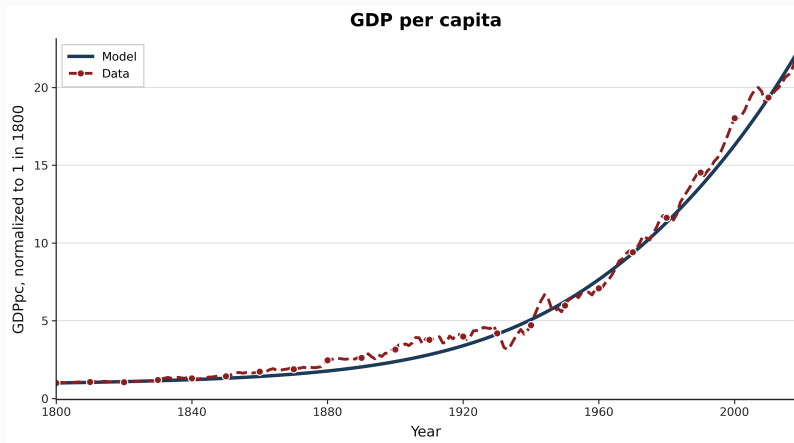


Figure 42: Calibration fit of targeted GDPPC path over 1850-2010. Data source: Gapminder.

Non Targeted Moments: 100\$ Spending Shock

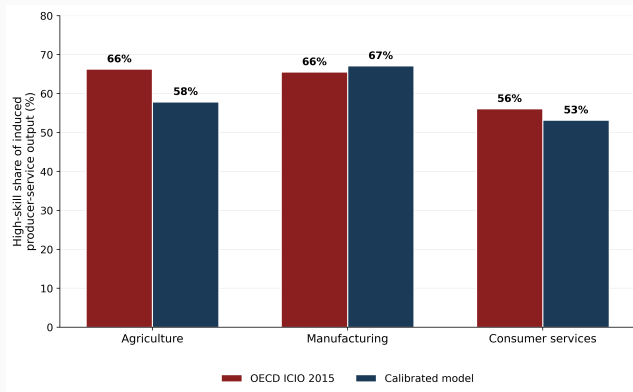


Figure 43: Bars show the high-skill share of producer-service output induced by a \$100 final-demand increase in each sector. The data exercise uses Leontief total requirements from the 2015 OECD ICIO table; the model exercise applies the same shock to the calibrated model economy. .

Stylized Fact 2: Sectoral College Shares over Development

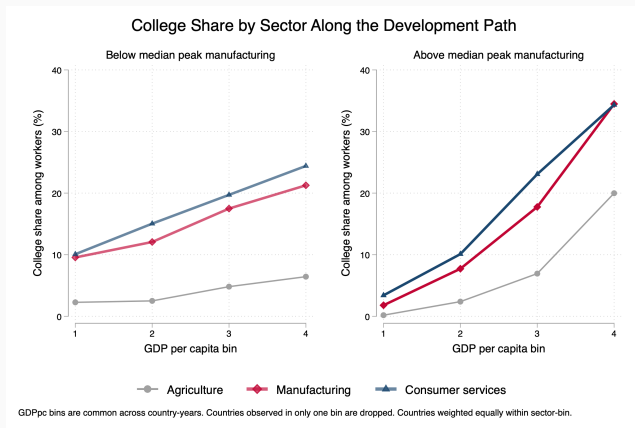


Figure 44: The figure plots average college employment shares by sector across GDP per capita bins, separately for countries below and above the median peak manufacturing employment share. Data source: IPUMS International (271 country-year samples).

Stylized Fact 2: Sectoral College Shares over Development

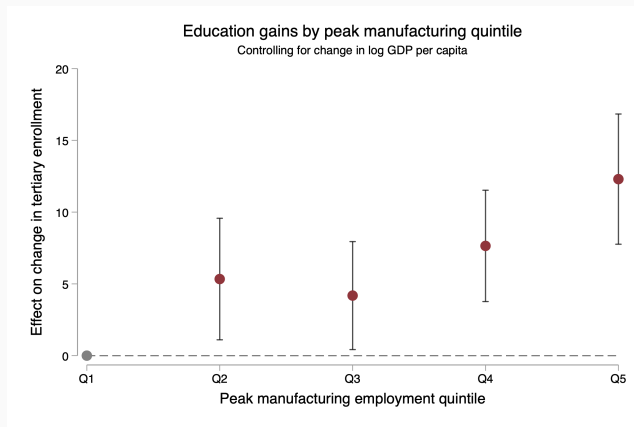


Figure 45: Education gains by peak manufacturing quintile. Coefficients are from a country-level long-difference regression of the change in tertiary enrollment on the change in log GDP per capita and indicators for peak manufacturing employment quintiles: $\Delta Educ_i = \alpha + \beta \Delta \log GDPpc_i + \sum_{q=2}^5 \delta_q 1\{PeakManufQuintile_i = q\} + \varepsilon_i$. The omitted group is the lowest peak-manufacturing quintile. Data sources: Barro-Lee (2015) and Gapminder.

Stylized Fact 3: Manufacturing and IO Linkages

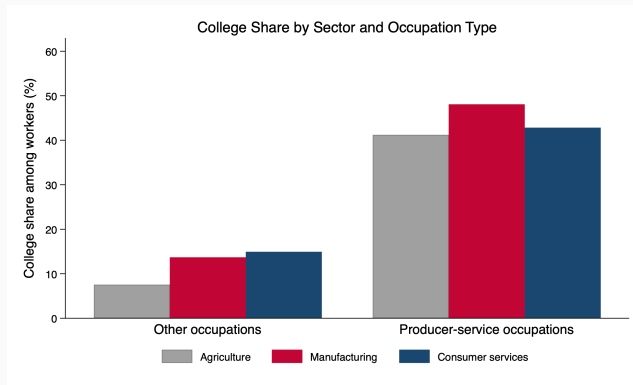


Figure 46: The figure shows college worker shares by broad sector, separately for other occupations and producer-service occupations. Data source: IPUMS International.

Stylized Fact 3: Manufacturing and IO Linkages Over Development

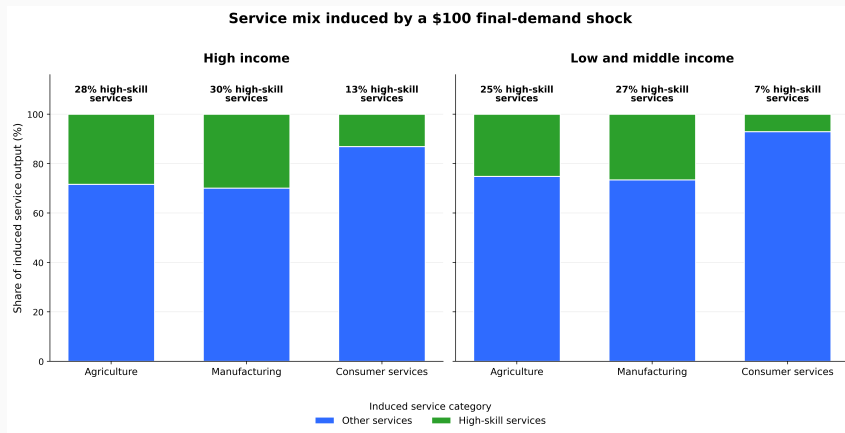


Figure 47: Manufacturing pulls toward high-skill services. Bars show the service-sector mix induced by a \$100 final-demand increase in agriculture, manufacturing, or consumer services, computed from Leontief total requirements. Data source: 2015 OECD Inter-Country Input-Output (ICIO)

Stylized Fact 3: Manufacturing and IO Linkages Final Expenditure Weights

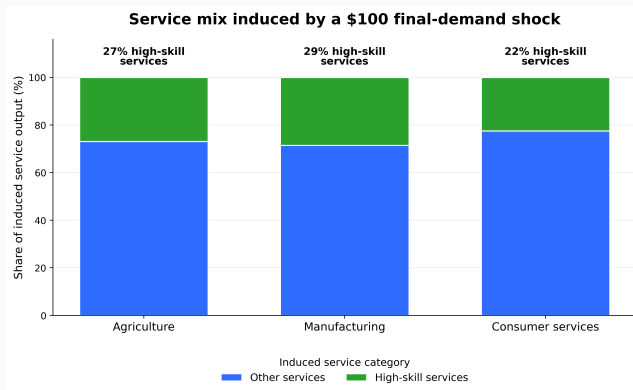


Figure 48: Bars show the service-sector mix induced by a \$100 final-demand increase in each sector, computed from Leontief total requirements. The consumer-services shock is allocated across all market services using observed final-consumption weights, allowing consumer purchases of finance, computer/information, and professional services. Data source: 2015 OECD ICIO.

Stylized Fact 3: Intermediate Shares of GO

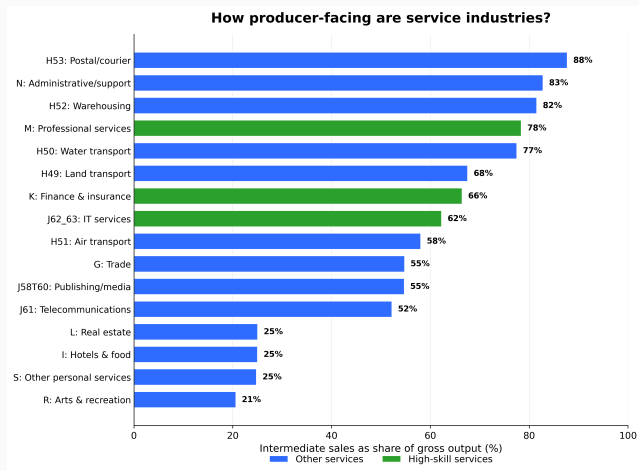


Figure 49: Bars show each service industry's intermediate sales as a share of gross output in the 2015 OECD ICIO table. Colors indicate the service taxonomy used in the main exercise.

Panel Data: Effects of % Man. Emp. during the Transition

Table 9: Panel Data (1850–1940)

	(1) Δ Fertility	(2) Δ Enrollment	(3) Δ Ed. Score	(4) Δ Fertility	(5) Δ Enrollment	(6) Δ Ed. Score
L.% Man Emp.	-0.282*** (0.067)	0.109*** (0.035)	6.267*** (0.504)	-0.312*** (0.064)	0.086*** (0.025)	2.181*** (0.567)
State × Year FE	No	No	No	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12758	12758	12748	12758	12758	12748
Within R2	0.022	0.060	0.075	0.522	0.546	0.568

Standard errors are clustered at the county level. Dependent variables are measured in first differences. Data source: US Census of Population 1850-1940.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

IV: Instrumenting the Peak

The geographic features which are conducive to manufacturing change over time. For example, proximity to coal, rivers, or the Great Lakes may matter more in some decades than others.

I hence construct an instrument by taking the fitted values from the following zero stage regression:

$$\widehat{\text{Peak \% Man Emp.}}_i = \sum_d \left(\widehat{\beta}_d^C \ln(\text{coal})_i + \widehat{\beta}_d^F \ln(\text{fall})_i + \widehat{\beta}_d^O \ln(\text{ohio})_i + \widehat{\beta}_d^{GL} \text{GreatLakes}_i \right) \cdot 1\{\text{peak decade} = d\}$$

- Each geographic characteristic is allowed to have a different effect by peak decade.
- Identification comes from time-varying relevance of geographic advantages.

First Stage Results

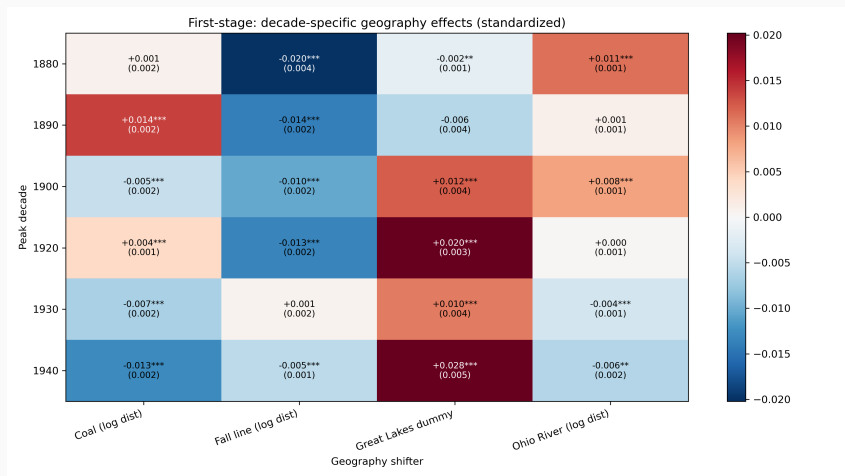


Figure 50: First stage (standardized) coefficients. Each geographic characteristic is interacted four decade peak dummy variables. Data source: US Census of population and Haines (2010).

DiD-IV: Effect on Educational Scores Within Services

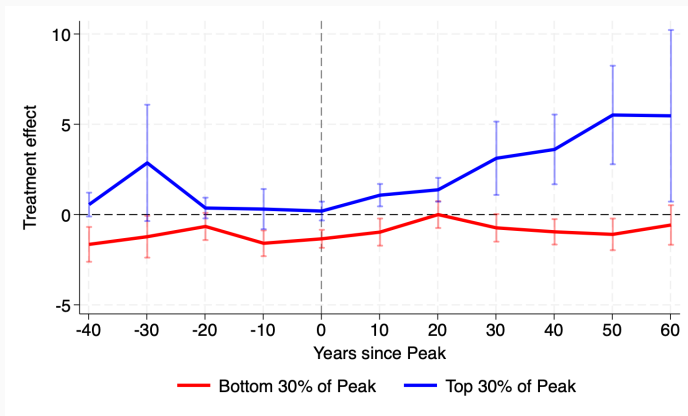


Figure 51: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k 1\{\text{Peak \% Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t ; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).